TSM 099 Arithmetic Test - Practice 1

Version 202610122020.16 | Name: _____

Complete and show all work on a separate sheet of paper. Record final answers on the answer sheet. Give exact answers and simplify completely.

Answer key follows the answer sheet. Attempt the test before checking it.

1. Evaluate the principal square root in each part.

(*a*) $\sqrt{25}$

(*b*) $\sqrt{-49}$

2. Complete the table by evaluating each side separately. State what the tested values establish; do not solve the equation.

$\frac{x^2-4}{x-2} = \frac{x^2+5x+6}{x+3}$

x	Left value	Right value	Equal?
6			
-6			
0			

3. Evaluate at (a) x = 2 and (b) x = -2. Show substitution.

$x^2 - 3x + 2$

4. Reduce each fraction to lowest terms.

(*a*) $\frac{14}{21}$

(*b*) $\frac{42}{66}$

5. Simplify the square root.

$\sqrt{18}$

Student answer sheet

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Question 1

(a) 10 pts

(b) 10 pts

Question 2

x = 6 4 pts

x = -6 4 pts

x = 0 4 pts

Conclusion 8 pts

Question 3

(a) 10 pts

(b) 10 pts

Question 4

(a) 10 pts

(b) 10 pts

Question 5

Radical 20 pts

Answer key

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Question 1

(a)	5	10 pts
(b)	7i	10 pts

Question 2

x = 6	Left 8; right 8; yes	4 pts
x = -6	Left -4; right -4; yes	4 pts
x = 0	Left 2; right 2; yes	4 pts
Conclusion	All tested values work; this alone does not prove an identity.	8 pts

Question 3

(a)	0	10 pts
(b)	12	10 pts

Question 4

(a)	$\frac{2}{3}$	10 pts
(b)	$\frac{7}{11}$	10 pts

Question 5

Radical	$3\sqrt{2}$	20 pts